Smart ICT Tool overview

Revision 1.0.1 from 14.04.2021

Programmiersysteme für die
Mikroelektronik GmbH
Südwestpark 100
D-90449 Nürnberg

Phone: +49 (0) 911-25 26 65-0
Fax: +49 (0) 911-25 26 65-66
Mail: info@promik.com
Web: http://www.promik.com

Certified according to:
- ISO 9001
- ISO 14001

Contents

ProMik

Programmiersysteme für die Mikroelektronik GmbH
Südwestpark 100
D-90449 Nürnberg

Phone: +49 (0) 911-25 26 65-0
Fax: +49 (0) 911-25 26 65-66
Mail: info@promik.com
Web: http://www.promik.com

Certified according to:
- ISO 9001
- ISO 14001

1 Overview

This document describes the way of handling the tool for determining the test coverage of different ODB++ related projects.

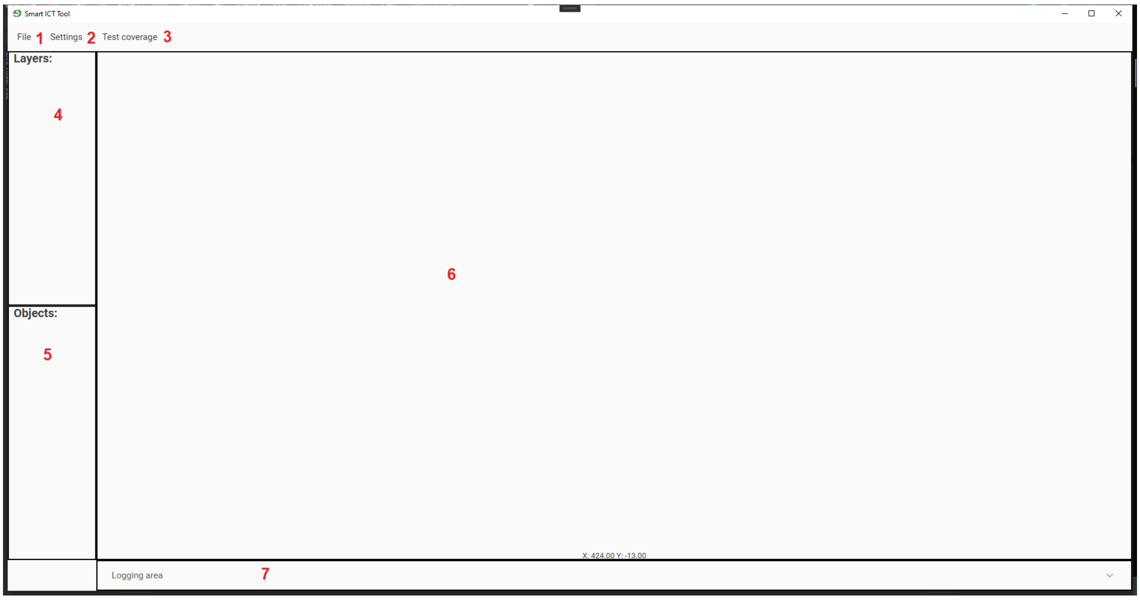
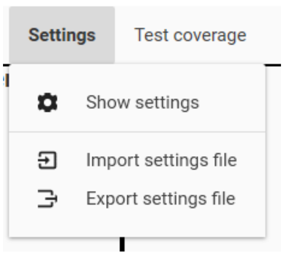


Figure 1: Tool overview

As shown in 1 the tool itself consits of seven different areas of interest:

1. File menu group - performing actions regarding the opening of project files, ...
2. Settings - selection of settings, importing and exporting them
3. Test coverage - all test coverage related actions
4. Layers - the layers overview
5. Objects - the test coverage objects list
6. the rendered component view
7. The logging area

2 Settings



Programmiersysteme für die Mikroelektronik GmbH
Südwestpark 100
D-90449 Nürnberg

Phone: +49 (0) 911-25 26 65-0
Fax: +49 (0) 911-25 26 65-66
Mail: info@promik.com
Web: http://www.promik.com

Certified according to:
- ISO 9001
- ISO 14001

Figure 2: Settings overview

As shown in 2 there are the following three menu items available:

1. Show settings - opens the settings page for all settings
2. Import settings file - a file chooser will pop up for selecting a settings database file to be imported
3. Export settings file - a file chooser will pop up for selecting the target file where the settings database should be exported to

2.1 Show settings

After clicking on the menu item All settings a new window will be shown containing all setting pages.

2.1.1 PCB Investigator API settings

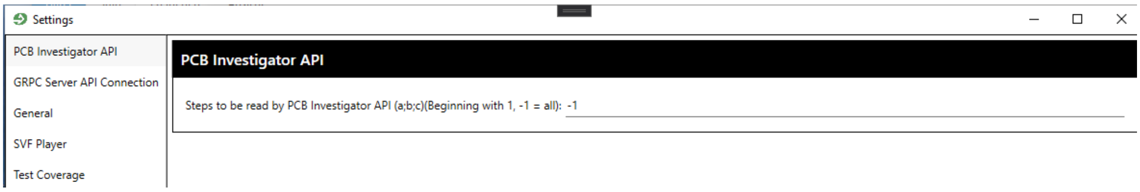


Figure 3: PCB Investigator API settings

As shown in 3 there is the possibility of changing the steps which should be considered for read out from the API. -1 means all steps should be read out. Other values could be defined semicolon separated

2.1.2 GRPC server API connection settings

Figure 4: GRPC server API connection settings

As shown in 4 there is the possibilty of changing the ip address of the GRPC server to connect to

ProMik

Programmiersysteme für die Mikroelektronik GmbH
Südwestpark 100
D-90449 Nürnberg

Phone: +49 (0) 911-25 26 65-0
Fax: +49 (0) 911-25 26 65-66
Mail: info@promik.com
Web: http://www.promik.com

Certified according to:
- ISO 9001
- ISO 14001

2.1.3 General settings

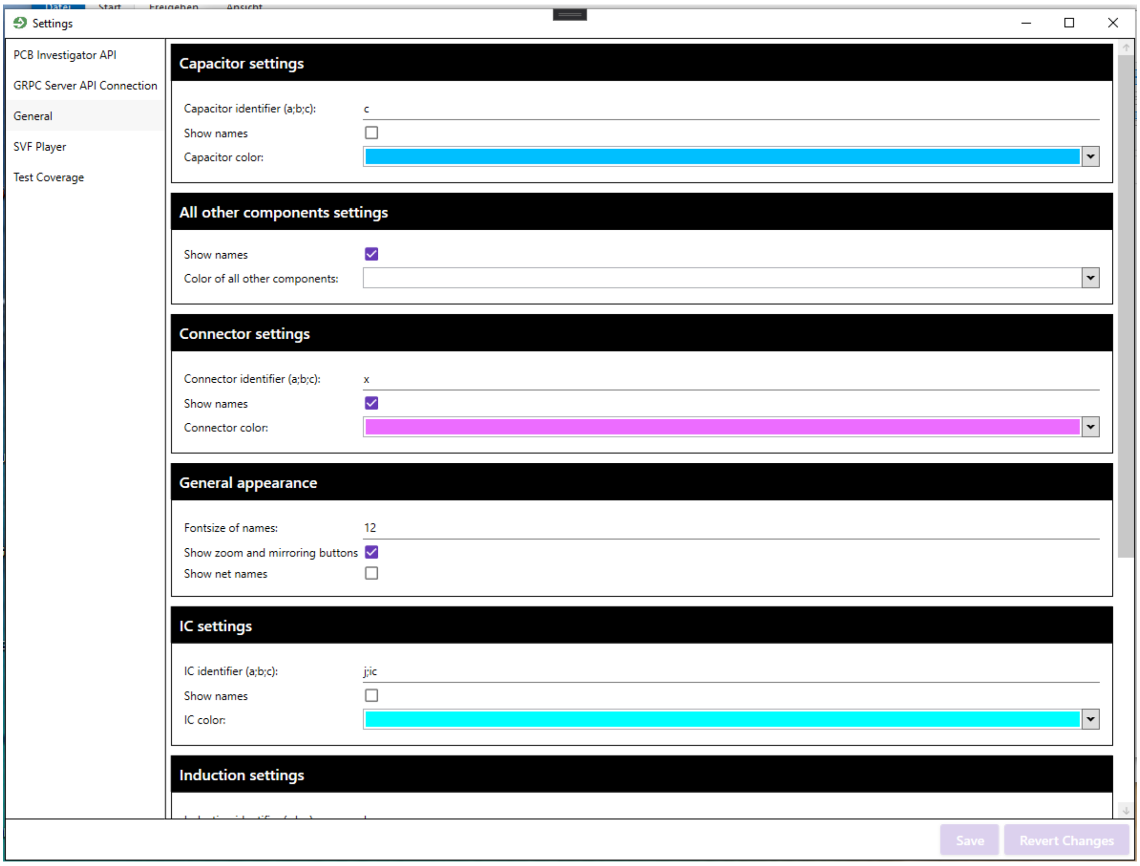


Figure 5: General settings 1

Within the first part of the general settings (5) there are the following settings available (identifier, show names and the color):

- Capacitor
- Other components
- Connector
- Ic

Fruthermore there are the following settings available:

- General appearance
  - Fontsize of names
  - Show zoom and mirroring buttons
  - Show net names

Programmiersysteme für die Mikroelektronik GmbH
Südwestpark 100
D-90449 Nürnberg

Phone: +49 (0) 911-25 26 65-0
Fax: +49 (0) 911-25 26 65-66
Mail: info@promik.com
Web: http://www.promik.com

Certified according to:
- ISO 9001
- ISO 14001

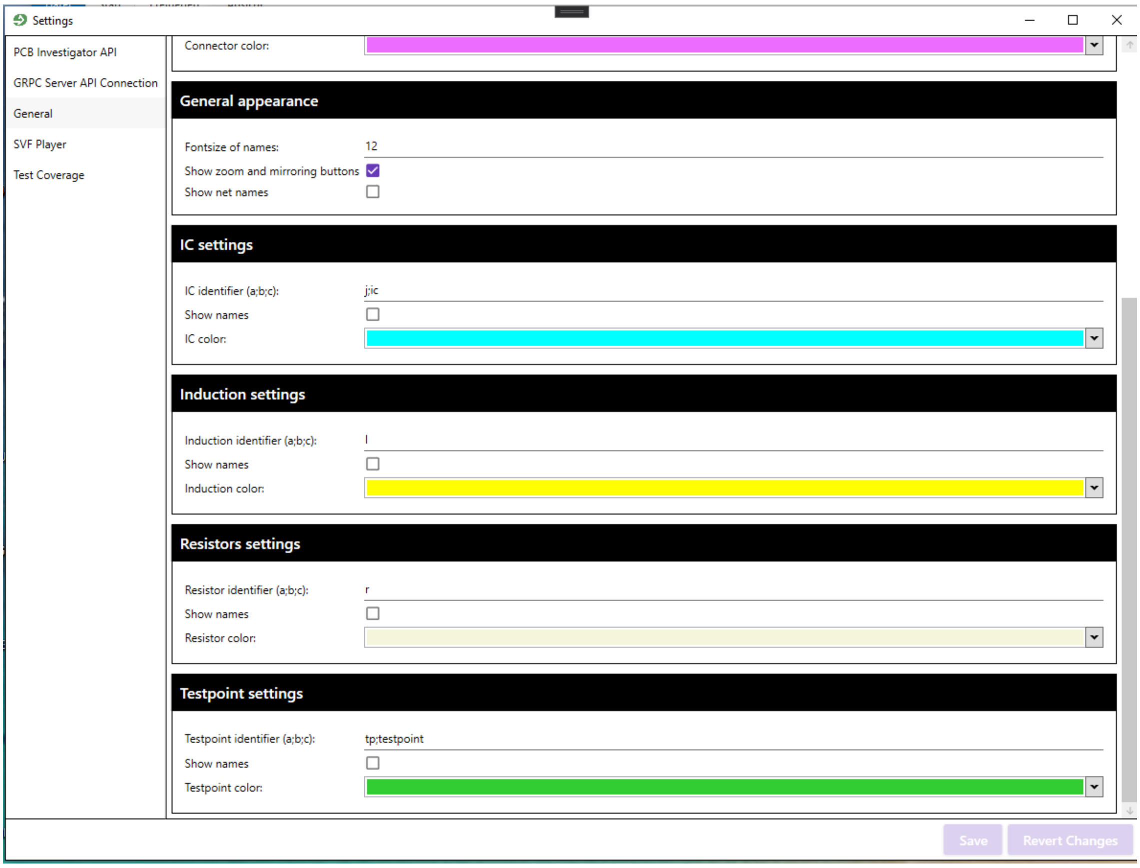


Figure 6: General settings 2

Within the second part of the general settings (6) there are the following settings available (identifier, show names and the color):

- Induction
- Resistor
- Testpoint

Programmiersysteme für die Mikroelektronik GmbH
Südwestpark 100
D-90449 Nürnberg

Phone: +49 (0) 911-25 26 65-0
Fax: +49 (0) 911-25 26 65-66
Mail: info@promik.com
Web: http://www.promik.com

Certified according to:
- ISO 9001
- ISO 14001

2.1.4 SVF Player settings

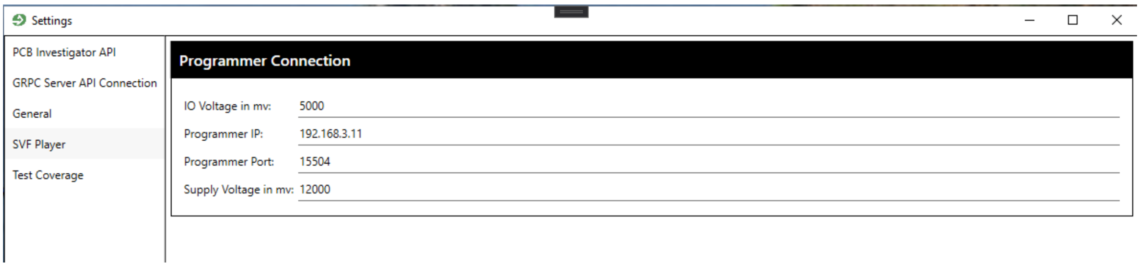


Figure 7: SVF Player settings

As shown in 7 there are the following settings available:

- IO Voltage in mv
- Programmer IP
- Programmer Port
- Supply Voltage in mv

2.1.5 Test Coverage settings

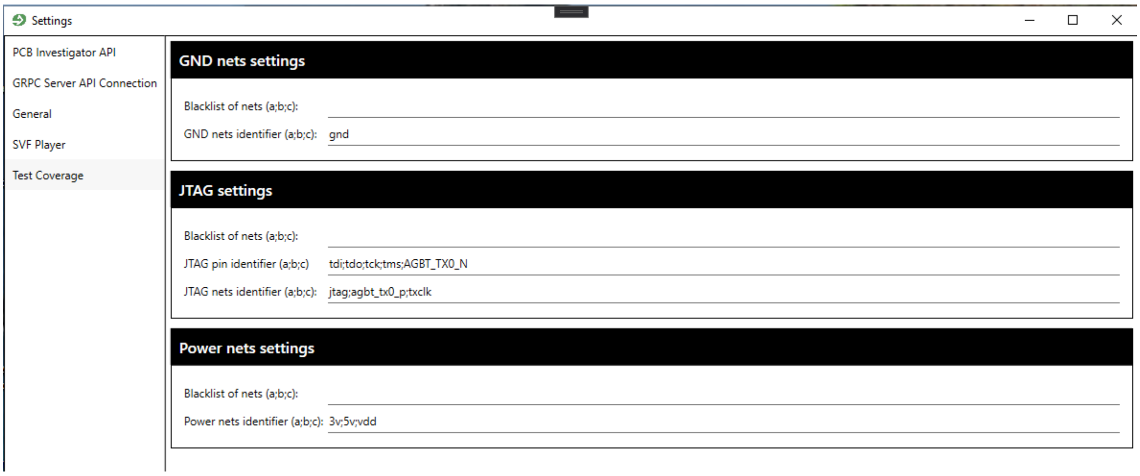


Figure 8: Test Coverage settings

As shown in 8 there are the following settings available (blacklist and identifier):

- Ground
- JTAG
- Power

ProMik
Programmiersysteme für die Mikroelektronik GmbH
Südwestpark 100
D-90449 Nürnberg

Phone: +49 (0) 911-25 26 65-0
Fax: +49 (0) 911-25 26 65-66
Mail: info@promik.com
Web: http://www.promik.com

Certified according to:
- ISO 9001
- ISO 14001

Further more there is the possibility of defining the JTAG pin identifier for making sure being able to use not just the default values **TMS, TDI, TDO, TCK**. That means in detail for correctly determining a JTAG device there should be the following requirements accomplished:

1. Set up the **IC identifier** setting from general settings in that way that the JTAG device is being recognized correctly as an IC component
2. Set up the **JTAG nets identifier** from the test coverage settings in that way that the JTAG device is correctly being recognized within a corresponding net
3. Set up the **JTAG pin identifier** setting from the test coverage settings in that way that the JTAG device is correctly being recognized by its pins

2.2 Import / Export settings file

By clicking on one of the menu items Settings -> Import settings file or Settings -> Export settings file afterward a new file selection dialog will be shown where to select the source / target file to be imported / exported. Further if the project file handling is enabled the settings file will automatically be stored within the project file and be loaded automatically on next opening of project file

3 Project file

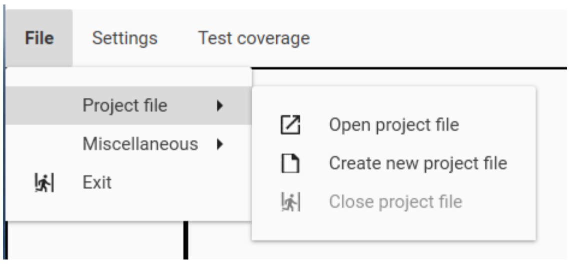


Figure 9: Project file handling

As shown in 9 there is the possibility of choosing the following menu items within the File -> Project file group:

- Open project file
- Create new project file

Programmiersysteme für die Mikroelektronik GmbH
Südwestpark 100
D-90449 Nürnberg
Phone: +49 (0) 911-25 26 65-0
Fax: +49 (0) 911-25 26 65-66
Mail: info@promik.com
Web: http://www.promik.com
Certified according to:
- ISO 9001
- ISO 14001

4 Miscellaneous

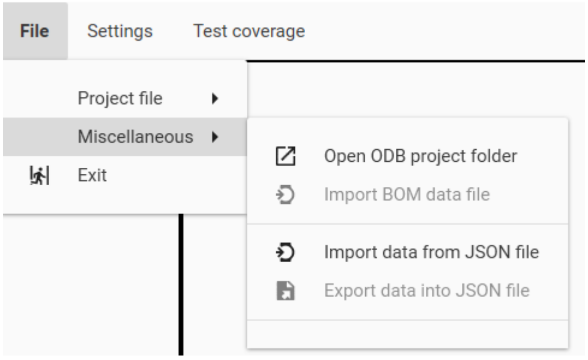


Figure 10: Miscellaneous menu items

As shown in 10 there are the following menu items available at File -> Miscellaneous:

- Open ODB project folder - after choosing that action a new dialog appears to select the folder which should be transferred to the GRPC server as zipped content for read out by using the PCB-Investigator API
- Import BOM data file - After any project was successfully loaded then the import of a valid bill of materials file will be available. After choosing this a new window appears where to set up all relevant settings for import
- Import data from JSON file - a new dialog appears where you are able to select any JSON formatted file containing previously exported data of this application
- Export data into JSON file - after any project was loaded successfully then this item will be accessible. After choosing it a new dialog appears where you could select the target for creation of a valid JSON formatted file containing all data

5 Creation of new project

To create a new project just first click on File -> Project file -> Create new project file. After ward there will be a dialog shown for selectin the target file to be saved at:

ProMik
Programmiersysteme für die Mikroelektronik GmbH
Südwestpark 100
D-90449 Nürnberg

Phone: +49 (0) 911-25 26 65-0
Fax: +49 (0) 911-25 26 65-66
Mail: info@promik.com
Web: http://www.promik.com

Certified according to:
- ISO 9001
- ISO 14001

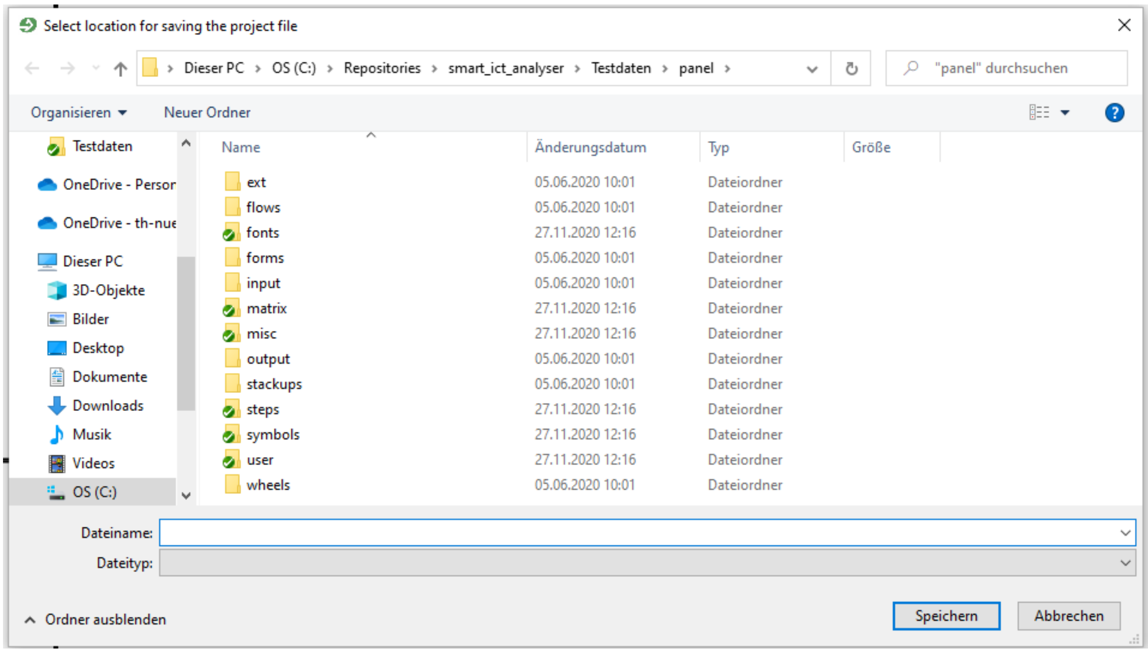


Figure 11: Project file creation dialog

After you confirmed your target within the dialog (11) then you could proceed to open a new project. For that make sure to have a valid connection to the GRPC server and click then File -> Miscellaneous -> Open ODB project folder. After you chose a program during the communication and parsing from the GRPC server there will be shown an animation:

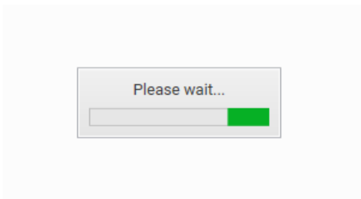


Figure 12: Please wait animation

After the process finished then if no values are defined for that components received a new dialog should be visible where to select the BOM file for importing:

ProMik
Programmiersysteme für die Mikroelektronik GmbH
Südwestpark 100
D-90449 Nürnberg
Phone: +49 (0) 911-25 26 65-0
Fax: +49 (0) 911-25 26 65-66
Mail: info@promik.com
Web: http://www.promik.com
Certified according to:
- ISO 9001
- ISO 14001

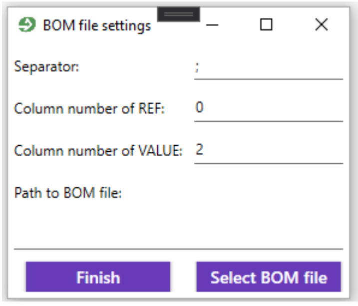


Figure 13: Select BOM file dialog

After you set up the settings:

- Separator
- Column number of REF
- Column number of VALUE
- Path to BOM file - could be set by clicking on Select BOM file where then a new dialog for selecting any file will be opened

within that dialog and selected a valid CSV file to be imported then proceed by clicking on finish.

After the process finished then on within the layers list there should be some layers available and the control buttons at the right bottom place will be visible:

Figure 14: Loaded project view

After selecting for example the first layer then all components which are included within that layer should be rendered within the component view.

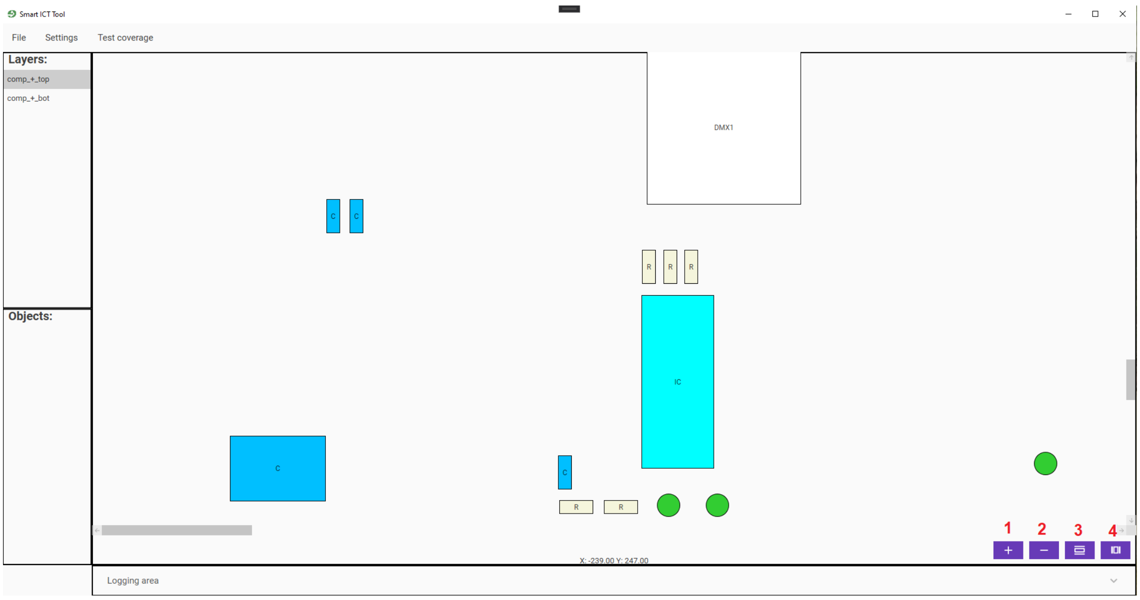


Figure 15: Compontens view for selected layer

As shown in 15 there is the possibility of scrolling with the mouse to move the whole view or just by clicking and dragging the whole content. Further there is the possibility of zooming in by pressing STRG and scrolling in or just by clicking on the zoom in button **1** and to zoom out by pressing STRG and scrolling out with the mouse or just by clicking the zoom out button **2**. Finally there is the possibility to mirror the whole view across the y axis **3** and the x axis **4**. Now to ensure not to have to use the GRPC server again the next time you will open this project click File -> Miscellaneous -> Export data into JSON file. After you chose it a new dialog will appear where you could type in just the name as the file itself will be stored within the project file as you are working within the project file mode. (Also all other project related data was automatically saved in the project file). Finally you should also export your current made settings by Settings -> Export settings file to make sure that the settings file is also included in the project file for later loading

6 Working with the components view

By right clicking on any object within the components view there will be a context menu visible:

ProMik®
Programmiersysteme für die Mikroelektronik GmbH
Südwestpark 100
D-90449 Nürnberg

Phone: +49 (0) 911-25 26 65-0
Fax: +49 (0) 911-25 26 65-66
Mail: info@promik.com
Web: http://www.promik.com

Certified according to:
- ISO 9001
- ISO 14001

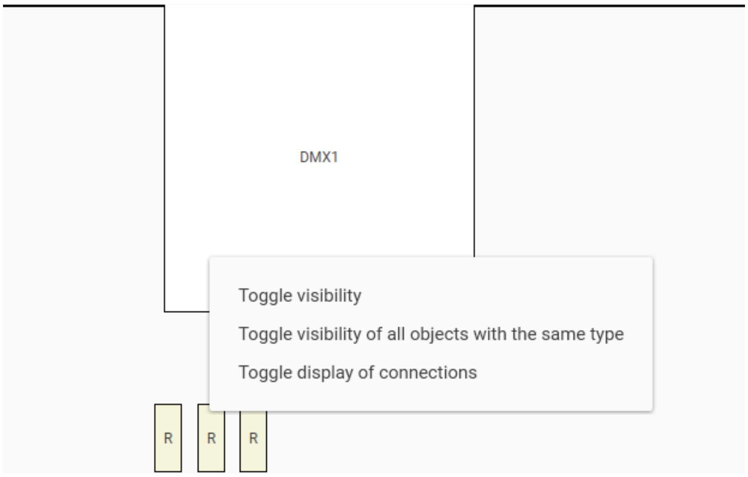


Figure 16: Context menu of right click on component

As shown at 16 you have the possibility of:

- Toggle visibility - if clicked then the according components transparency will be set to high
- Toggle visibility of all objects with the same type - if clicked then all the according components which have the same type as the selected ones transparency will be set to high
- Toggle display of connections - if clicked then all connections of the component will be shown as plain lines

6.1 Toggle visibility

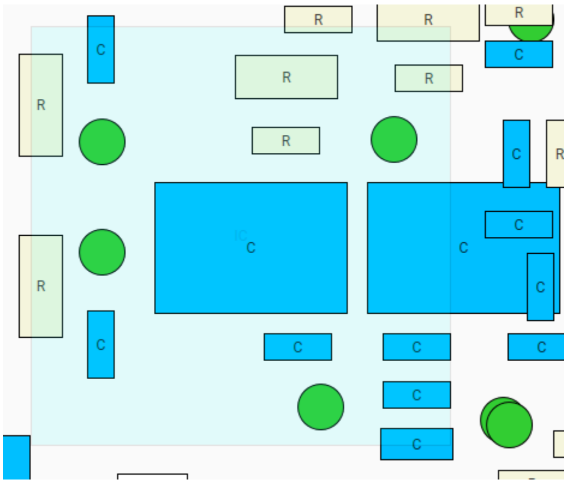


Figure 17: Toggle visibility

As shown in 17 after the visibility was toggled then the underlining objects could be determined

ProMik®
Programmiersysteme für die Mikroelektronik GmbH
Südwestpark 100
D-90449 Nürnberg

Phone: +49 (0) 911-25 26 65-0
Fax: +49 (0) 911-25 26 65-66
Mail: info@promik.com
Web: http://www.promik.com

Certified according to:
- ISO 9001
- ISO 14001

6.2 Toggle visibility of all objects with the same type

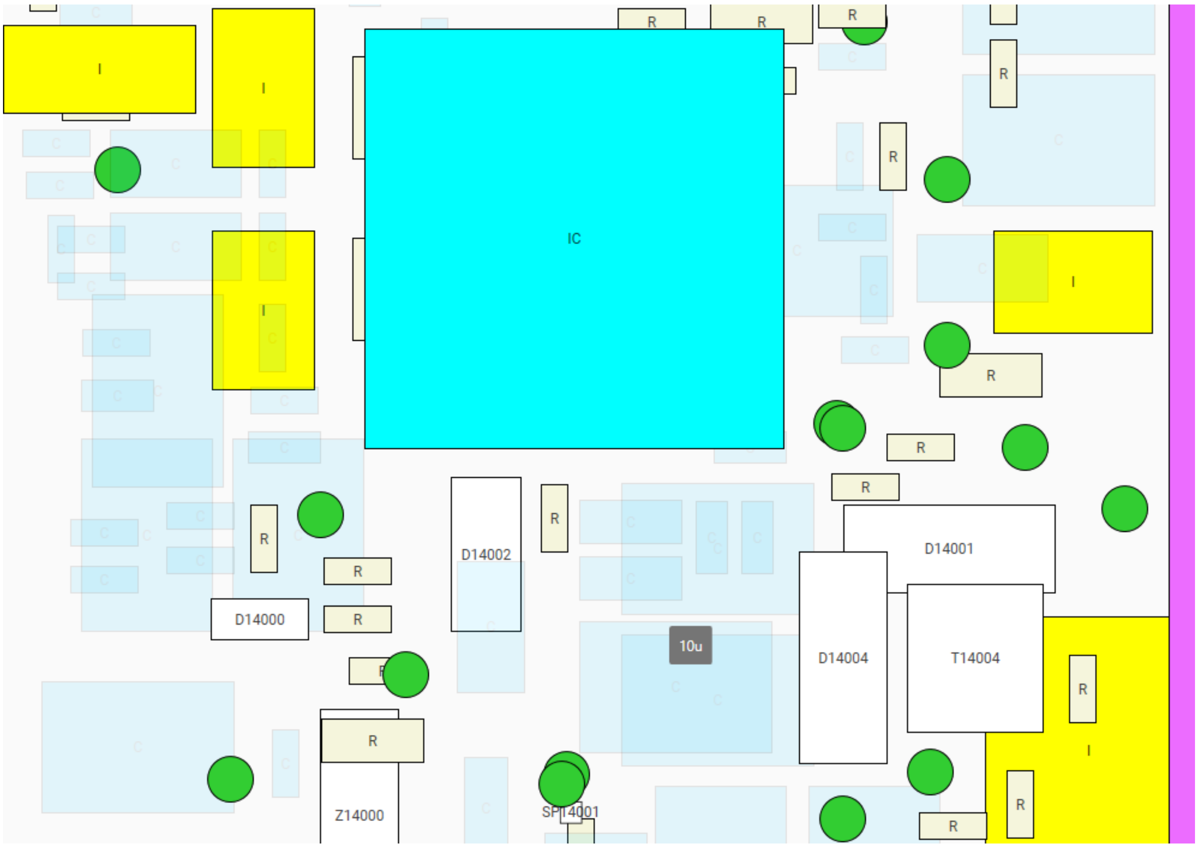


Figure 18: Toggle visibility of all objects with the same type

As shown in 17 after the visibility was toggled then the underlining objects of all toggled objects of the same type could be determined

ProMik
Programmiersysteme für die Mikroelektronik GmbH
Südwestpark 100
D-90449 Nürnberg

Phone: +49 (0) 911-25 26 65-0
Fax: +49 (0) 911-25 26 65-66
Mail: info@promik.com
Web: http://www.promik.com

Certified according to:
- ISO 9001
- ISO 14001

6.3 Toggle display of connections

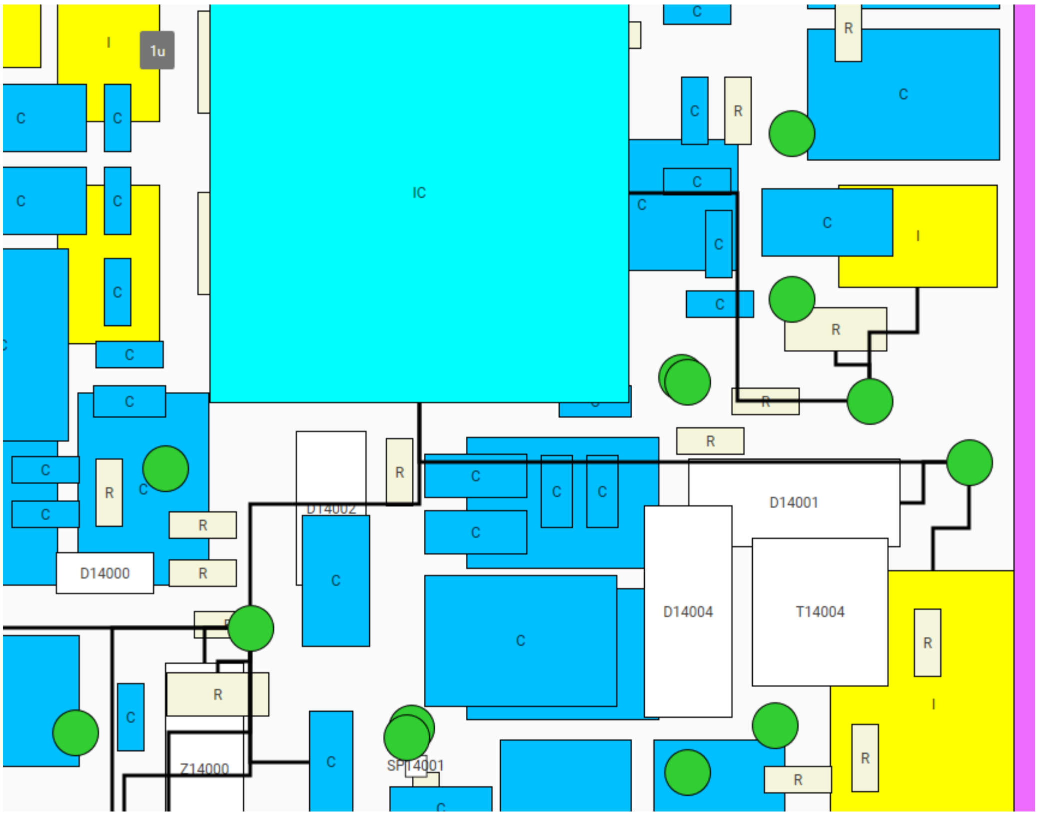


Figure 19: Toggle display of connections

As shown in 19 after the connections were toggled then all lines will be drawn between the connecting objects. Further there is the possibility to hover over any connection to see the following information:

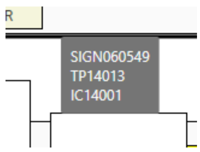


Figure 20: Legend of connection

As shown in 20 there will be the following information available:

- Net name
- Selected component for toggling the connections
- Connected component by this line to the selected component

Programmiersysteme für die Mikroelektronik GmbH
Südwestpark 100
D-90449 Nürnberg

Phone: +49 (0) 911-25 26 65-0
Fax: +49 (0) 911-25 26 65-66
Mail: info@promik.com
Web: http://www.promik.com

Certified according to:
- ISO 9001
- ISO 14001

6.4 Test coverage

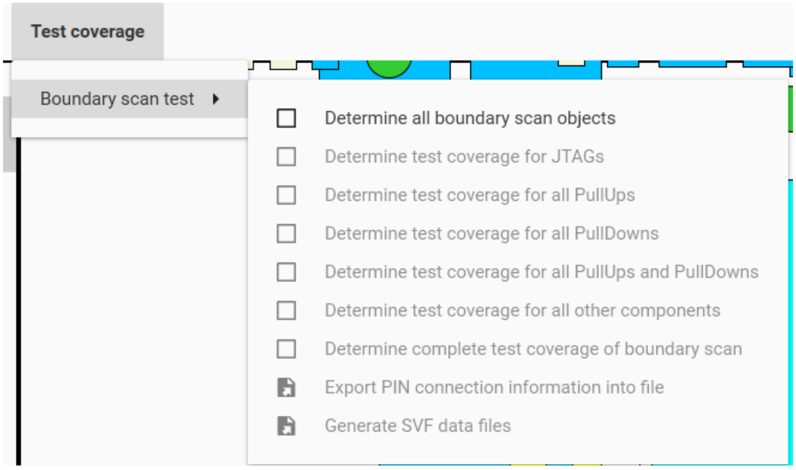


Figure 21: Test coverage menu group

As shown in 21 within the Test coverage -> Boundary scan test -> menu group there are the following menu items available:

- Determine all boundary scan objects - before being able to perform any other test related action you have first to choose this one. (Settings will be considered)
- Determine test coverage for JTAGs
- Determine test coverage for all PullUps
- Determine test coverage for all PullDowns
- Determine test coverage for all PullUps and PullDowns
- Determine test coverage for all other components
- Determine complete test coverage of boundary scan
- Export PIN connection information into file
- Generate SVF data files

6.5 Determine all boundary scan objects

After you selected Test coverage -> Boundary scan test -> Determine all boundary scan objects then the objects list should be filled with new selection possibilities:

ProMik
Programmiersysteme für die Mikroelektronik GmbH
Südwestpark 100
D-90449 Nürnberg

Phone: +49 (0) 911-25 26 65-0
Fax: +49 (0) 911-25 26 65-66
Mail: info@promik.com
Web: http://www.promik.com

Certified according to:
- ISO 9001
- ISO 14001

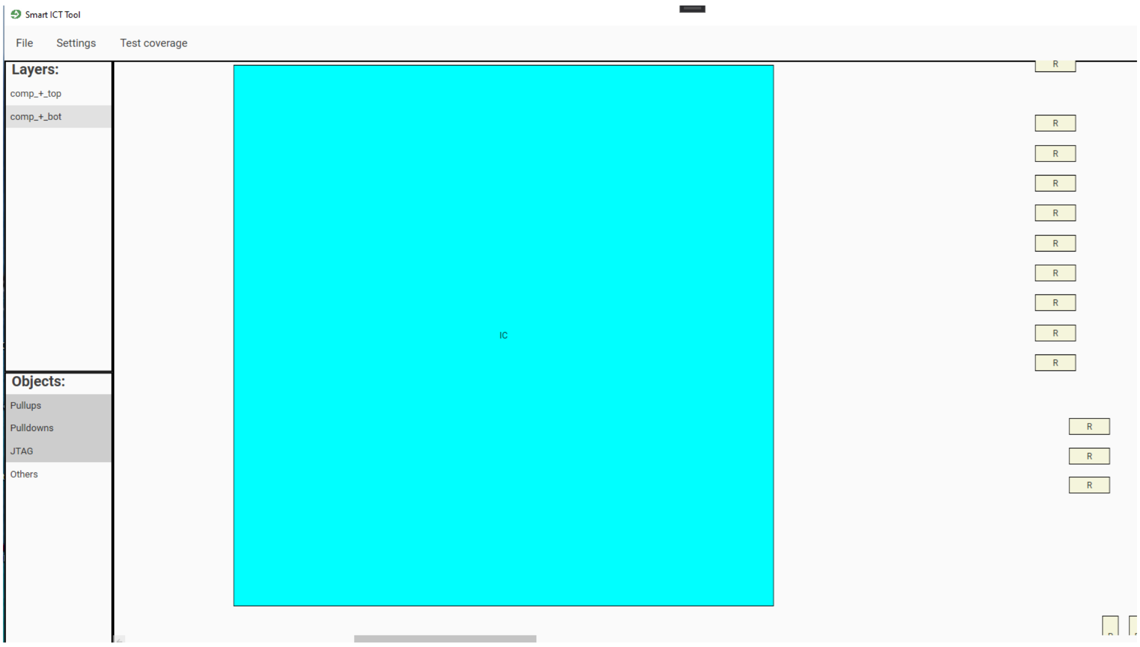


Figure 22: Determine all boundary scan objects

Now you could (22) select just the test coverage related objects of interest to be shown within the component view.

6.6 Performing any test coverage of objects

After you click for example on Test coverage -> Boundary scan test -> Determine test coverage for all PullUps then the calculated value will be shown as text and further more each connection to a according pull up resistor will be shown in green color:

ProMik
Programmiersysteme für die Mikroelektronik GmbH
Südwestpark 100
D-90449 Nürnberg
Phone: +49 (0) 911-25 26 65-0
Fax: +49 (0) 911-25 26 65-66
Mail: info@promik.com
Web: http://www.promik.com
Certified according to:
- ISO 9001
- ISO 14001

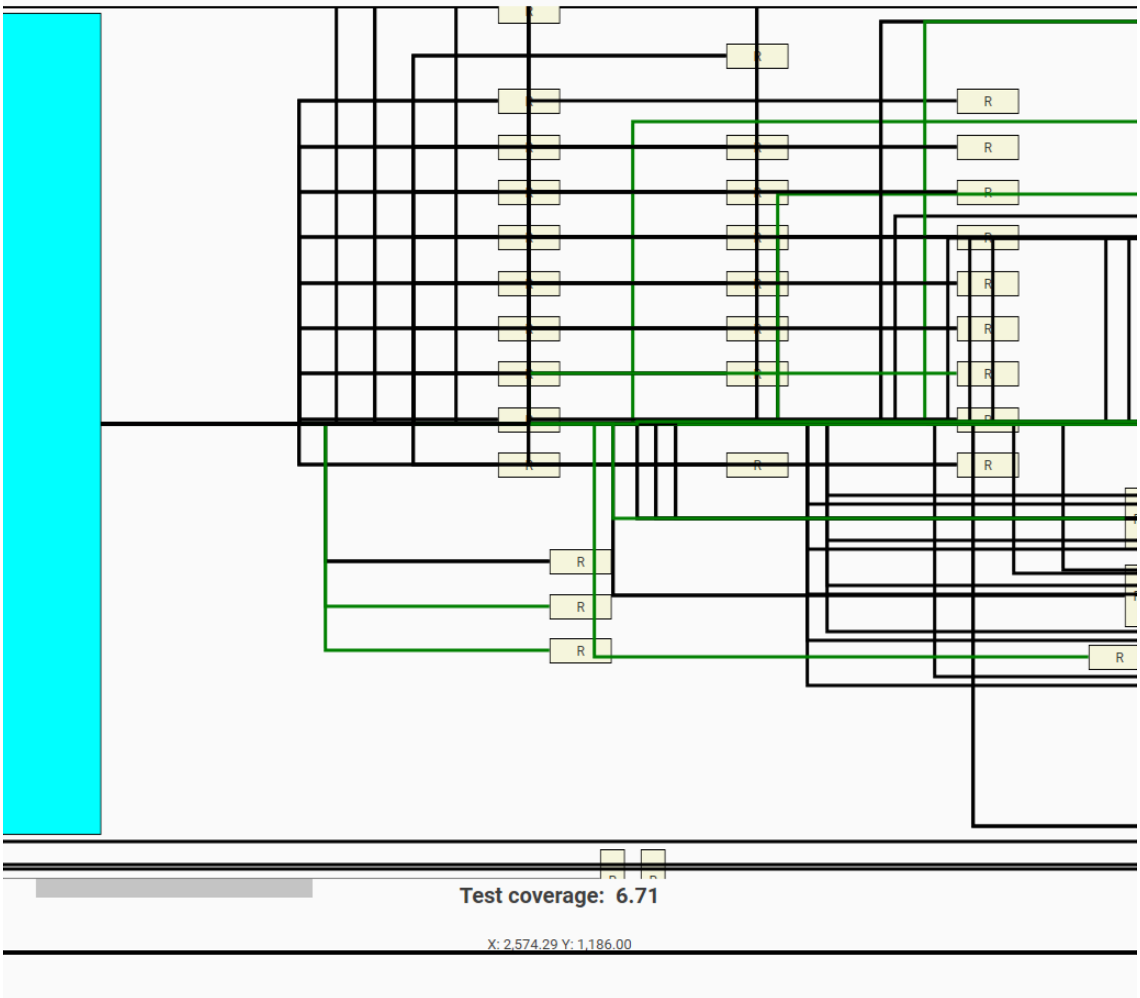


Figure 23: Determine test coverage for all PullUps

If you click again on Test coverage -> Boundary scan test -> Determine test coverage for all PullUps then the according checkbox will be unchecked and every connection is been showed just in black color again. This kind of procedure is also available for all other operations to detect the test coverage.

6.7 Export PIN connection information into file

After you click on Test coverage -> Boundary scan test -> Export PIN connection information into file a new dialog should appear where you can select the destination of file saving. After you selected that then the according file should be generate containing all pin information of interest of all JTAG devices of the loaded project.

Programmiersysteme für die
Mikroelektronik GmbH
Südwestpark 100
D-90449 Nürnberg

Phone: +49 (0) 911-25 26 65-0
Fax: +49 (0) 911-25 26 65-66
Mail: info@promik.com
Web: http://www.promik.com

Certified according to:
- ISO 9001
- ISO 14001

6.8 Generate SVF data files

If you click on Test coverage -> Boundary scan test -> Generate SVF data files then first a dialog will be shown to select the target folder where to save the created files at. This folder will only be considered if you are not working in the project file mode. If so the files will be automatically been saved within the project file. Afterward you will be asked for selecting the accoring BSDL file for all JTAG devices found. After you selected the BSDL file you will be informed about to make sure that the programmer is correctly connected to the JTAG device as the default vector first needs to be read out:

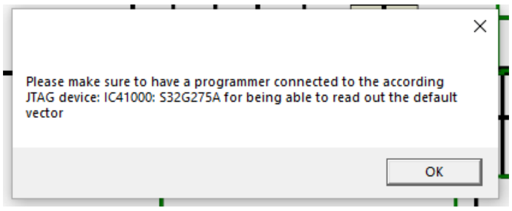
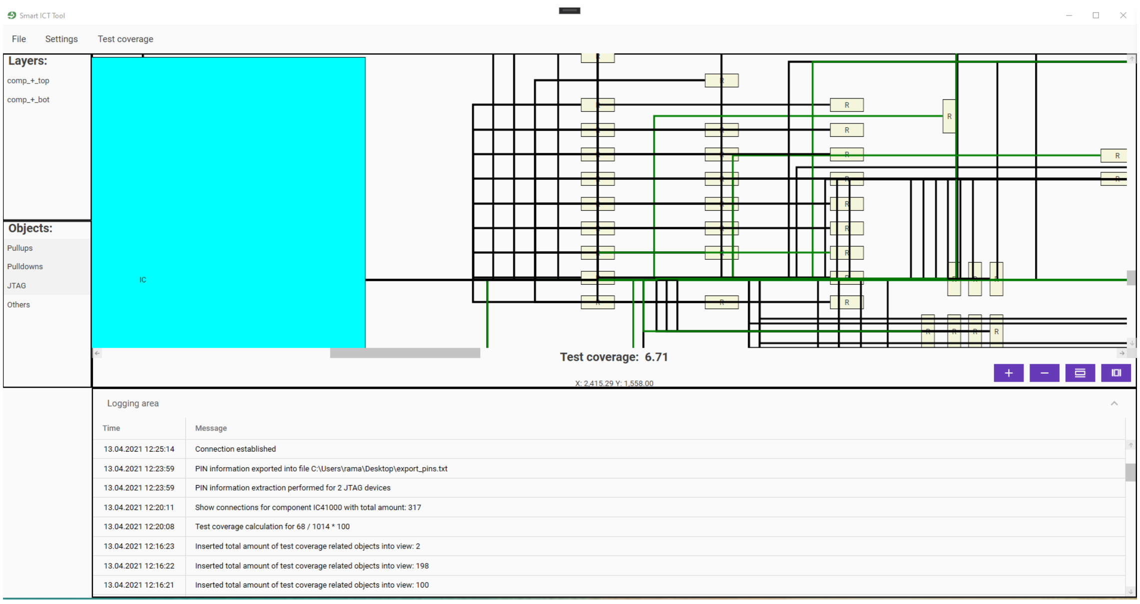


Figure 24: Make sure to have a programmer connected

After you confirmed the dialog 24 then the read out will be performed and the according SVF files for playing the pull down and pull up resistors will be generated.

7 Log panel

By clicking on the expander which is located at the very bottom of the application the log panel will be visible. It contains every single message logged during the working with the tool:



ProMik
Programmiersysteme für die Mikroelektronik GmbH
Südwestpark 100
D-90449 Nürnberg

Phone: +49 (0) 911-25 26 65-0
Fax: +49 (0) 911-25 26 65-66
Mail: info@promik.com
Web: http://www.promik.com

Certified according to:
- ISO 9001
- ISO 14001

Figure 25: Logging area

8 Drag and drop within the application

You are further able to drag and drop directly files or folders to the application area. If something valid was dropped then it automatically should be opended or imported and if project mode is enabled saved further within the project file

Programmiersysteme für die Mikroelektronik GmbH
Südwestpark 100
D-90449 Nürnberg

Phone: +49 (0) 911-25 26 65-0
Fax: +49 (0) 911-25 26 65-66
Mail: info@promik.com
Web: http://www.promik.com

Certified according to:
- ISO 9001
- ISO 14001

9 Acroyms

PullUps - Pull up resistors which are directly connected to a power supply PullDowns - Pull down resistors which are directly connected to ground connection JTAG - Joint Test Action Group (A connection device with the possibility of advanced electrical testing mechanisms) IC - Integrated circuit (A complex component consisting of many connections and other internal components)

ProMik®
Programmiersysteme für die Mikroelektronik GmbH
Südwestpark 100
D-90449 Nürnberg

Phone: +49 (0) 911-25 26 65-0
Fax: +49 (0) 911-25 26 65-66
Mail: info@promik.com
Web: http://www.promik.com

Certified according to:
- ISO 9001
- ISO 14001

Revision History

Revision	Date	Author(s)	Description
1.0.0	10.02.2021	RAM	Initial version of documentation
1.0.1	14.04.2021	RAM	Overworked complete content by using new style of tool

Programmiersysteme für die
Mikroelektronik GmbH
Südwestpark 100
D-90449 Nürnberg

Phone: +49 (0) 911-25 26 65-0
Fax: +49 (0) 911-25 26 65-66
Mail: info@promik.com
Web: http://www.promik.com

Certified according to:
- ISO 9001
- ISO 14001